MEMORY DIVERGENCE IS NOT BEHAVIORAL DIVERGENCE: STAGE-RESOLVED TRANSPORT IN SELF-IMPROVING AGENT MEMORY

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ABSTRACT

Persistent-memory agents are often evaluated as if changing what they store implied changing how they later behave. We show that this inference can fail even under a controlled state intervention. Using the released ReasoningBank writer, we hold a source trajectory fixed and flip only the terminal success/failure reflection branch. Reward-conditioned writing is robust: all 20 complete Shopping pairs produce different persistent memories, and on a disjoint eight-trajectory control the between-branch memory distance exceeds a stronger semantic-preserving same-mode wording perturbation by 0.105 (exact $p = 0.0078$). Yet the effect attenuates sharply across the native reuse pipeline. A forced fixed-evidence memory injection can produce terminal sensitivity (mean absolute success-rate difference 0.15625, permutation $p = 0.00074$), showing that the paired states can matter when directly supplied. Under native Shopping retrieval, however, branch-specific memory is exposed on 125/172 held-out retrieval opportunities while the terminal success difference falls to 0.02083 ($p = 0.4289$; 34/36 cells are zero), and the first-action total-variation difference is 0.06944 ($p = 0.5801$; 0/36 modal actions differ). A second-domain Reddit replication again shows robust write divergence (4/4) but sparse, sign-heterogeneous native terminal transport (mean absolute difference 0.125, $p = 0.2253$; 6/8 cells are zero). These results identify a stage-resolved boundary: *persistent-state divergence is not behavioral divergence*. Reward-conditioned memory is a strong upstream state intervention, but realized behavioral authority is filtered by retrieval exposure and policy uptake. We therefore argue that self-improving memory systems should be evaluated as a transport chain—write → exposure → uptake → outcome—rather than by memory distance or endpoint performance alone.

1 INTRODUCTION

Self-improving agents increasingly convert completed episodes into persistent natural-language memory. This creates an update channel that sits between ordinary evaluation and parameter learning: a terminal label can decide what durable state the agent carries into later tasks. Existing work has studied unreliable judges and reward models (Tan et al., 2025; Lambert et al., 2024), outcome-conditioned reflection (Shinn et al., 2023; Ouyang et al., 2026), learned memory construction (Wang et al., 2025; Li et al., 2026), and reward-dependent memory valuation or reuse (Asadolahi et al., 2026; Yang et al., 2026). A missing question is more basic: *when a reward-conditioned writer changes persistent state, how much of that state difference actually survives the later memory pipeline and becomes behavior?*

This distinction matters because current evaluation practice often collapses several different objects. A memory representation may change strongly while never being retrieved. A retrieved branch-specific memory may enter context without changing the next action. A changed action may still leave the terminal task outcome unchanged. Conversely, a forced-memory test may reveal that two memories are behaviorally capable of producing different outcomes even though the native retrieval-and-uptake pipeline rarely realizes that leverage. Treating any one of these measurements as a proxy for the others can misdiagnose both the mechanism and the appropriate intervention.

We study this boundary in the released ReasoningBank implementation. Its writer exposes a particularly clean intervention surface: score one selects a success-reflection writer and score zero selects a failure-reflection writer over the same trajectory. We hold the source trajectory fixed and flip only this reward-conditioned writer branch. This isolates an operational write-time state contrast without learning a new construction policy. A disjoint same-mode semantic-rewording control then tests the strongest simple reduction—that the observed state difference is merely ordinary prompt sensitivity.

The upstream effect is robust. Across 20 complete Shopping success/failure writer pairs, the persistent memories diverge. In the frozen eight-trajectory prompt control, reward-mode memory divergence is 0.700 versus 0.595 under stronger semantic-preserving same-mode wording perturbations; the paired excess is 0.105 with exact $p = 0.0078$. Thus the reward/reflection branch is a reliable state intervention at write time.

The downstream result is not a monotone continuation of that effect. When paired memories are directly forced into a fixed-evidence future-task packet, they are capable of changing terminal behavior: the fully crossed 4-by-4 confirmatory experiment yields mean absolute success-rate difference 0.15625 with permutation $p = 0.00074$. But this is a *leverage* test, not a native-pipeline transport estimate. Under native Shopping reuse, branch-specific memory is retrieved on 125/172 held-out opportunities, yet the mean absolute terminal success difference is only 0.02083 ($p = 0.4289$), with 34/36 cells exactly zero. The corresponding first-action distribution has TV distance 0.06944 ($p = 0.5801$), and no frozen state changes its modal first action across reward branches. A Reddit replication again changes memory in 4/4 paired writes, but native terminal transport remains sparse and sign-heterogeneous: mean absolute difference 0.125, $p = 0.2253$, with 6/8 cells zero and opposite signs in the two nonzero cells.

These measurements change the scientific object. The paper is not a claim that reward error uniformly contaminates future behavior, nor a claim that memory distance is a behavioral metric. Instead, it identifies a *stage-resolved transport boundary*:

$$\text{reward label} \to \text{durable write} \to \text{retrieval exposure} \to \text{policy uptake} \to \text{outcome}. \tag{1}$$

The treatment is strong at the first stage, demonstrably has downstream leverage when forced, but is sharply attenuated under native exposure and uptake. The mismatch between these measurements is itself the central finding.

Contributions. We make three tightly coupled contributions. First, we formulate reward-conditioned memory as a controlled persistent-state intervention by holding the source trajectory fixed and varying only the released success/failure writer branch; a stronger same-mode wording control separates this intervention from generic prompt sensitivity. Second, we introduce a stage-resolved evaluation that distinguishes *state divergence*, *forced behavioral leverage*, *native retrieval exposure*, *policy uptake*, and *terminal transport*; the experiments show that these quantities can differ by an order of magnitude and should not be used interchangeably. Third, we replicate the write/transport separation across Shopping and Reddit and derive a practical evaluation rule: persistent-memory systems should localize where an update's effect attenuates before assigning it behavioral authority or designing a repair.

Closest-work boundary. Prior work already establishes that rewards can shape memory construction, valuation, or reuse. We surrender those broad claims. Our residual contribution is measurement and identification: under a fixed writer policy and same-trajectory intervention, durable branch-specific state is easy to create, while native behavioral transport is sparse and domain/task dependent. This is distinct from learning a better memory policy or proposing another memory gate. Indeed, our attempted evidence-gated method extension stopped before behavioral evaluation because its semantic-validity signal could not be independently qualified; we therefore keep the present paper centered on the measured transport boundary rather than manufacturing a solution after seeing the results.

2 A Stage-Resolved Model of Persistent-State Transport

Let a task x produce trajectory τ and let the terminal evaluator emit $r \in \{0, 1\}$. A reward-conditioned memory writer constructs

$$M_r = G(\tau, r). \tag{2}$$

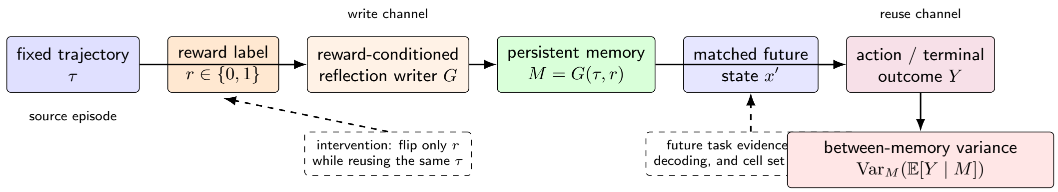

Figure 1: The stage-resolved reward-to-memory transport chain. The source trajectory is fixed while the terminal reward label selects the released reflection writer. The resulting state must then survive retrieval exposure and policy uptake before it can affect terminal outcome. A forced-memory packet bypasses the native exposure stage and therefore measures downstream *leverage*, not native end-to-end transport.

In the released ReasoningBank implementation, $r = 1$ selects a success-reflection objective and $r = 0$ selects a failure-reflection objective. Our paired intervention holds τ fixed and changes only the selected writer branch. The first estimand is therefore a persistent-state contrast,

$$D_M(\tau) = d(M_1, M_0)\,, \tag{3}$$

where d is a prespecified memory-distance statistic. This quantity answers whether the update surface changes. It does *not* answer whether the changed state is later exposed to the policy or behaviorally consequential.

We therefore decompose downstream transport into three additional stages. For a held-out future state x', let $E_r(x') \in \{0,1\}$ indicate whether branch-specific memory from M_r is retrieved into the future context. Let $A_r(x')$ denote the first structured policy action under that context, and let $Y_r(x')$ denote the terminal task outcome. The stage-specific objects are

$$T_E = \Pr(E_r = 1), \tag{4}$$
$$T_A = d_A(P(A \mid M_1, x'), P(A \mid M_0, x'))\,, \tag{5}$$
$$T_Y = |\mathbb{E}[Y \mid M_1, x'] - \mathbb{E}[Y \mid M_0, x']|\,. \tag{6}$$

A large D_M can coexist with a small T_A or T_Y if branch-specific content is rarely retrieved, ignored after retrieval, dominated by the current observation, or behaviorally redundant. This is the core identification point of the paper.

2.1 Forced leverage versus native transport

A direct memory injection and a native memory-bank rollout estimate different quantities. In a forced packet, the experimenter supplies M_r to the policy, so exposure is mechanically set to one. The terminal contrast

$$L_Y(x') = |\mathbb{E}[Y \mid do(M = M_1), x'] - \mathbb{E}[Y \mid do(M = M_0), x']| \tag{7}$$

asks whether the paired states are *capable* of changing behavior under matched evidence. In native reuse, retrieval remains part of the system, and the end-to-end contrast combines exposure and uptake. Consequently, $L_Y > 0$ does not imply a large native T_Y.

This distinction resolves an apparent contradiction in our experiments. Forced injection produces a substantial terminal contrast, while native Shopping and Reddit transport is sparse. Rather than choosing one result as the "real" effect, the stage model treats both as measurements of different links in the same causal chain.

2.2 Attenuation as the mechanism diagnostic

For descriptive comparison, define a stage attenuation ratio whenever the upstream denominator is nonzero,

$$\alpha_{j \to k} = \frac{T_k}{T_j}. \tag{8}$$

We do not interpret this ratio as a universal causal coefficient because the stage metrics use different scales. Its role is diagnostic: it forces the analysis to ask where a large upstream intervention stops being behaviorally expressed. The scientific claim rests on the separately measured stage endpoints and their qualitative ordering, not on a single composite score.

The resulting evaluation principle is simple: *do not infer behavioral authority from persistent-state divergence alone*. An update should be characterized at the write, exposure, uptake, and outcome stages separately, with forced-injection tests used only to establish latent leverage when native transport is weak.

3 Stage 1: Reward-Conditioned Writing Robustly Changes Persistent State

We first test the write stage with paired interventions over released WebArena Shopping trajectories. Within each pair, the task text and full action history are fixed before the writer is called. One branch uses the released success-reflection objective and the other uses the released failure-reflection objective; the resolved writer model is matched within pair and temperature is zero. Raw responses and the pre-writer trajectory projection are content-addressed before analysis.

The initial F0 tranche requested six source trajectories. Four returned complete memory outputs in both conditions; the two incomplete requests terminated in the failure-label writer before assistant text existed and were excluded from the paired content estimand. Among the four complete pairs, every pair changed normalized memory content and the extracted memory-item title set, with mean token-set Jaccard distance 0.735. Receipt-level forensics preserve the two incomplete arms as provider-response failures rather than treating them as negative memory examples.

The later frozen breadth tranche expands the write-stage evidence to 20 complete Shopping success/failure pairs. All 20/20 pairs produce different branch memories, with pooled token-set Jaccard distance 0.673. When combined with the four Reddit pairs used in the cross-domain replication, the D0 lineage contains 24/24 paired source experiences with the same pre-writer trajectory projection, the same resolved writer model within pair, temperature zero, and different branch-memory content. This establishes a robust operational branch contrast at the persistent-state level.

The content changes are not only token-level. A deterministic post-ready strategy-slot audit maps frozen memory titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification slots. In the original four complete F0 pairs, the strategy-slot set changes in all four. We treat this as a structural diagnostic rather than a semantic-validity claim: the current data do not certify each textual delta atom as a pure reward-caused proposition because explicit decoding seeds or same-condition noise-floor replications are unavailable.

The write-stage result therefore supports a narrow statement: under the released writer policy, flipping the reward-conditioned reflection branch on the same source experience reliably produces a different durable state. It does not by itself establish that the difference will be retrieved, followed, or reflected in terminal behavior. Section 4 first tests the strongest simple prompt-sensitivity reduction; Section 5 then measures downstream transport stage by stage.

4 Ruling Out Generic Prompt Sensitivity

The reward intervention changes the reflection instruction together with the reward semantics. A generic prompt-sensitivity account therefore predicts that a sufficiently strong wording perturbation inside one reflection mode should produce comparable memory divergence. We test that reduction on eight released trajectories disjoint from F0 and balanced across original outcomes.

Before provider calls, we freeze semantic-preserving rewordings for both reflection modes. Under token-set Jaccard distance, the released success-versus-failure prompts differ by 0.114. The within-success rewording differs by 0.171 and the within-failure rewording by 0.152. The control thus perturbs surface wording more strongly while preserving the reflection objective.

Each fresh trajectory is written with the released success mode and the released failure mode. The same writer also processes one frozen semantic-preserving rewording of each mode at temperature

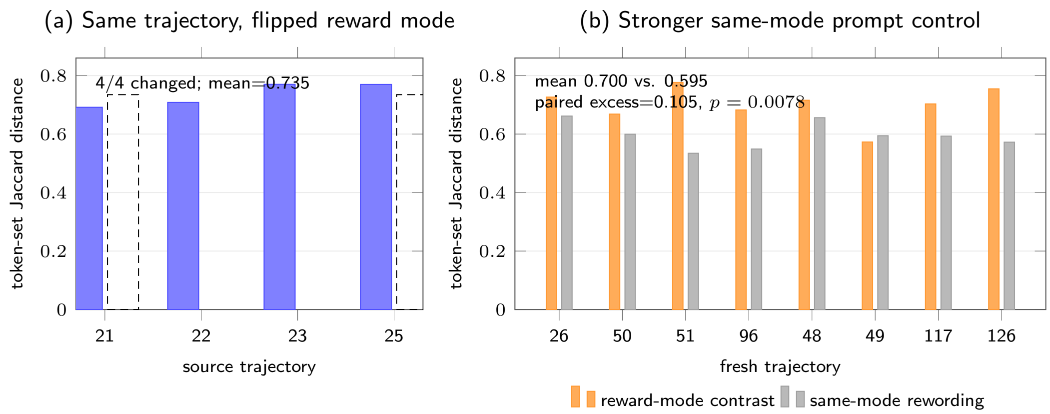

Figure 2: Write-channel evidence and its strongest simple control. (a) Among the four F0 pairs complete in both writer conditions, flipping only the reward-conditioned writer mode changes every memory. (b) On eight fresh control trajectories, reward-mode memory divergence exceeds stronger semantic-preserving same-mode prompt rewording. The control mean is 0.595 versus 0.700 between reward modes; paired excess is 0.105 with exact $p = 0.0078$.

zero. Let B_i be the memory distance between the two released reward modes and let W_i be the mean within-mode rewording distance. The frozen paired statistic is

$$\Delta_{\text{prompt}} = \frac{1}{8}\sum_{i=1}^{8}(B_i - W_i). \tag{9}$$

All 32 control outputs complete. Mean reward-mode distance is 0.700, while the stronger same-mode wording control has mean 0.595. The paired excess is 0.105, and an exact one-sided sign-flip test gives $p = 0.0078$. This passes the frozen 0.10 effect margin together with the $p < 0.05$ gate. The reward-conditioned write effect therefore survives a same-information prompt-wording reduction that is deliberately stronger in lexical perturbation.

5 Stages 2–4: Where Does the Write Effect Survive?

The write intervention is scientifically interesting only if we distinguish its downstream routes. We therefore evaluate three questions separately: (i) can the paired memories affect behavior when directly supplied; (ii) are branch-specific memories actually exposed under the native memory pipeline; and (iii) once exposed, do they produce detectable policy or terminal differences?

5.1 Forced injection establishes latent behavioral leverage

We first bypass native retrieval and directly inject each paired memory into a fixed future-task packet. This experiment answers a capability question: are the two persistent states behaviorally distinguishable when exposure is forced to one?

The original fixed-state probe contains 12 aligned success-memory versus failure-memory rollouts; 7/12 choose different structured next actions. We then freeze all four complete initial F0 source-memory pairs and four future tasks before terminal-policy execution. Every source pair is crossed with every future task, giving 16 cells. The confirmatory run uses eight fresh rollouts per memory condition in every cell, for 256 completed calls. Mean absolute terminal success-rate difference is 0.15625, above the frozen 0.15 practical floor, and the 100,000-draw within-cell permutation test gives $p = 0.00074$.

This positive result is deliberately interpreted as *forced leverage*. The paired states can change downstream outcomes under matched evidence when the experimenter makes them available. The cell effects are heterogeneous: eight cells are zero, six have positive failure-minus-success effects, and two have negative effects. The two largest cells account for 78.2% of squared-effect mass. Thus even the forced experiment does not support a uniform harmful direction.

5.2 Native Shopping retrieval exposes memory but strongly attenuates branch transport

We next restore the native memory-bank pathway. After expanding the frozen Shopping memory bank, branch-specific source memories are retrieved on 125 of 172 held-out retrieval opportunities. This rules out the simplest explanation for weak native effects—that the memories never enter future context at all. Exposure is substantial.

Despite that exposure, native terminal transport is small. Across the frozen 36 Shopping branch-comparison cells, mean absolute success difference is 0.02083 with permutation $p = 0.4289$; 34/36 cells are exactly zero. The first-action stage is similarly weak: mean total-variation distance between reward branches is 0.06944 with $p = 0.5801$, and 0/36 frozen states change their modal first action. A post-hoc working-memory diagnostic yields branch shift 0.00335 with $p = 0.2052$. Together these measurements localize the dominant attenuation after retrieval exposure and before stable policy uptake.

This is the central mechanism result. The same reward-conditioned update produces a robust durable-state contrast, and the same paired states have measurable leverage under forced injection, yet native reuse rarely converts the branch-specific difference into a distinct first action or terminal outcome. Persistent-state mutation and behavioral authority are therefore empirically separable.

5.3 Outcome-blind control and second-domain replication

An outcome-blind structured-memory control produces a small but statistically detectable terminal contrast of 0.04514 ($p = 0.0048$), below the frozen 0.15 practical floor, with 32/36 cells zero. This control is useful because it shows why statistical significance alone is not the scientific endpoint: sparse tiny effects can pass a null test without approaching the magnitude observed under forced branch leverage.

We then repeat the write-versus-native-transport measurement on Reddit. Reward-conditioned writing again diverges in 4/4 paired source memories. Native terminal transport is larger than Shopping in magnitude but remains unresolved and highly sparse: mean absolute success difference is 0.125 with $p = 0.2253$, 6/8 cells are zero, and the two nonzero cells have opposite signs. The cross-domain result therefore supports heterogeneity rather than a universal attenuation constant or uniformly harmful reward-error effect.

5.4 Why the old one-step interpretation is wrong

A one-step story would summarize the 256 forced rollouts as evidence that "reward error propagates to future behavior." The native measurements show why that statement is too coarse. Forced injection estimates the effect conditional on making a particular memory state available; native transport additionally depends on retrieval competition, context composition, and policy uptake. The difference between 0.15625 forced terminal leverage and 0.02083 native Shopping terminal transport is therefore not an inconsistency to average away. It identifies where the system filters the update.

The practical consequence is an evaluation protocol rather than a new memory algorithm. For any persistent update, report at least one state-change metric and one downstream uptake/outcome metric, and localize the first stage where the intervention attenuates, vanishes, or changes sign. This prevents a large representation difference from being mistaken for a large behavioral effect.

6 Related Work

Reward and judge reliability. JudgeBench and RewardBench establish that LLM judges and reward models can be unreliable before their outputs are used for learning (Tan et al., 2025; Lambert et al., 2024); Plan-RewardBench shows additional brittleness on tool-integrated trajectories (Wang et al., 2026). Those works evaluate the scoring surface. We ask what happens after such a score controls a persistent state update.

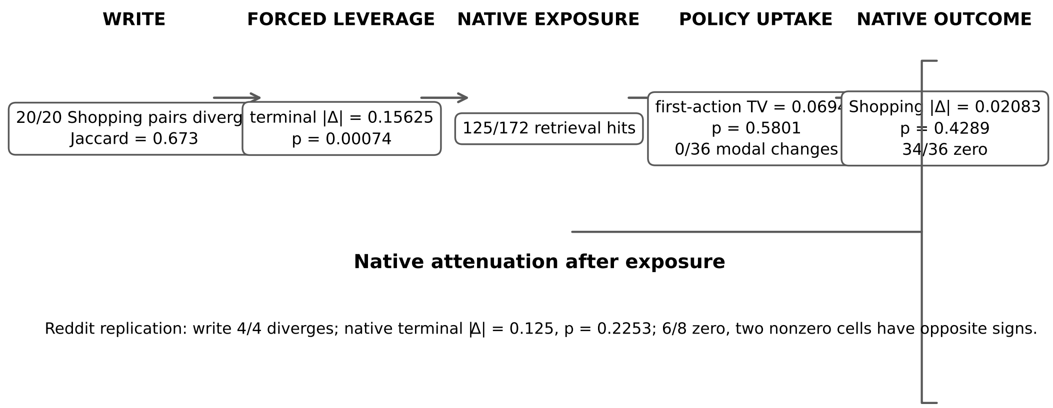

Figure 3: Stage-resolved transport boundary. Reward-conditioned writing is robust, and forced injection demonstrates that paired states can have downstream leverage. Under the native Shopping pipeline, retrieval exposure remains substantial but branch-specific first-action and terminal effects collapse; Reddit again shows sparse, sign-heterogeneous native transport. Values are kept on their native scales and should not be read as a single monotone effect-size axis.

Outcome-conditioned and learned memory construction. ReasoningBank explicitly distills different memories from successful and failed trajectories (Ouyang et al., 2026), Reflexion stores evaluative verbal feedback for later trials (Shinn et al., 2023), and Agent Workflow Memory extracts reusable procedures from interaction histories (Wang et al., 2024). Mem-α learns how to construct external memory from downstream reward (Wang et al., 2025), while AttriMem augments outcome reward with attribution-derived process feedback (Li et al., 2026). We therefore do not claim feedback-driven memory construction as new. Our intervention freezes the writer policy and uses the success/failure branch only as a controlled state-update surface.

Memory valuation, reuse, and memory-reward feedback. Memory Reward Inflation studies how imperfect proxy reward can be amplified as memories are scored and reused (Asadolahi et al., 2026). RoMeRL studies misleading trajectory-level utility updates and the memory-reward trap (Yang et al., 2026). These works place the key failure in valuation or later reuse. Our measurement begins earlier, at write-time state construction, but its main contribution is not merely moving the failure upstream: it measures how much of that upstream state contrast survives native retrieval and policy uptake.

Variance and longitudinal fragility. Ye et al. show run-to-run variance, curriculum sensitivity, and memory underspecification in self-improving agents (Ye et al., 2026). Their results motivate measuring downstream instability, but variance alone does not localize where an update enters the system. Our forced/native contrast separates latent behavioral leverage from end-to-end native transport and shows that the largest attenuation can occur after a persistent state has already been created.

Evaluation boundary. The closest conceptual distinction is therefore not another memory architecture but an evaluation decomposition. Retrieval success is not policy uptake; policy uptake is not terminal effect; and a forced-memory experiment bypasses native exposure. The paper's residual claim is that these stages are empirically non-equivalent for reward-conditioned persistent memory. This is why we report strong write divergence, positive forced leverage, substantial native exposure, and weak native first-action/terminal transport together rather than selecting only the positive endpoint.

7 LIMITATIONS

Our intervention identifies an operational success-versus-failure writer contrast, not atom-level causal purity for every generated sentence. The original incomplete failure-label provider calls remain excluded from paired text estimands because no assistant text exists for those arms. More broadly, the current frozen pool has no explicit decoding-seed binding or same-condition same-trajectory noise-floor replication for each memory atom. We therefore avoid claiming that every textual delta is individually caused by the reward bit.

The stage measurements also use different experimental surfaces. Forced fixed-evidence injection deliberately bypasses native retrieval and should be interpreted only as latent behavioral leverage. Native Shopping and Reddit estimates include the system's retrieval and policy-uptake pathway. Their numerical magnitudes should not be combined into a single causal coefficient. Source-faithful live WebArena browser navigation remains unavailable in the released environment, so the native experiments use the frozen supported pathway rather than a newly reconstructed live endpoint.

The native transport effects are heterogeneous and sparse. In Shopping, 34/36 terminal cells are zero; in Reddit, 6/8 are zero and the two nonzero cells have opposite signs. The data therefore do not support a universal directional effect, a universal attenuation factor, or a model-independent effect size. The current writer/task evidence is also limited in model and domain breadth.

Finally, our attempted evidence-gated residual method extension did not reach behavioral evaluation. It stopped at method realization because the frozen support contained no independently qualified outcome-independent semantic-validity adjudicator. This is not evidence that the method effect is negative, nor does it invalidate the stage-resolved measurements. We report this boundary to avoid turning a qualification failure into an unrun method claim.

8 CONCLUSION

Reward-conditioned writing is a strong persistent-state intervention, but persistent-state divergence is not the same object as behavioral divergence. Under same-trajectory success/failure reflection, Shopping memories diverge robustly, and the effect exceeds a stronger same-mode wording control. When paired memories are forced into matched future-task packets, they can produce substantial terminal sensitivity. Yet under native reuse, the effect sharply attenuates: retrieval exposure remains substantial while first-action and terminal branch differences become sparse, small, and domain dependent.

This stage separation changes how persistent-memory self-improvement should be evaluated. A memory update should not be credited with behavioral authority merely because its representation changes, nor should a forced-memory effect be presented as native end-to-end transport. The relevant scientific object is the chain

$$\text{write} \rightarrow \text{exposure} \rightarrow \text{uptake} \rightarrow \text{outcome},$$

and the key diagnostic is the first stage at which a controlled state intervention attenuates or changes sign.

For the released ReasoningBank-based configuration, the strongest conclusion is therefore a measurement boundary rather than a new memory algorithm: reward labels can reliably alter durable state, while the native system often filters that branch-specific state before it becomes stable behavior. This distinction provides a concrete evaluation principle for self-improving agents and a sharper target for future mechanism or intervention work.

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A F0 Reproduction Details

A.1 Released trajectory sample

The F0 sample uses six task IDs from the released AWM shuffle-seed-42 Shopping parquet: 21 through 25, plus 47. The deterministic sampler selects the lowest task IDs with parseable trajectory records within each original-outcome stratum. Original failures are 21 through 22 plus 24. Original successes are 23 plus 25 and 47.

A.2 Memory construction intervention

Each released trajectory is summarized into an action trace. The same trace is inserted into two prompts. One prompt contains the released ReasoningBank success-reflection system instruction. The paired prompt contains the released failure-reflection system instruction. The task prompt stays fixed. The writer model stays fixed. Temperature stays at zero. The raw provider output is archived before metric computation.

A.3 Pair-level F0 results

Table 1: Complete paired interventions.

Task	Original outcome	Content changed	Token distance
21	failure	yes	0.691489
22	failure	yes	0.708108
23	success	yes	0.769953
25	success	yes	0.769608

The extracted memory-item title set changes for every complete pair. The mean token-set Jaccard distance is 0.734789. A deterministic post-ready strategy-slot audit maps the frozen titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification/completeness. All four complete pairs change their slot set. The slot-set Jaccard distances are 0.571, 0.400, 0.500, and 0.500 for tasks 21, 22, 23, and 25, respectively, with mean 0.493. This is a structural diagnostic over frozen titles, not an embedding-based semantic claim.

Provider failures for task 24 and task 47 are excluded from the paired analysis. Both are failure-label calls with `ArkResponseStateError`; neither produces assistant text or a raw memory SHA. Private response identifiers are retained and GET-only recovery finds no text. The four completed failure-label calls produce 777, 852, 1,113, and 1,296 output tokens. These receipts distinguish provider response-state failures from a later memory-parser rejection, while leaving arm-dependent completion bias unresolved.

B PROMPT-PERTURBATION CONTROL DETAILS

The control set is disjoint from the F0 requests. Each fresh trajectory is written under both released reward modes. One semantic-preserving rewording is also evaluated within each mode. Table 2 reports the paired distances used in the frozen sign-flip test.

Table 2: Full fresh prompt-perturbation control. Between is the released success-versus-failure memory distance; Within averages the two same-mode rewording distances.

Task	Original outcome	Between	Within	$B_i - W_i$
26	failure	0.726	0.662	0.065
50	failure	0.668	0.599	0.069
51	failure	0.776	0.534	0.242
96	failure	0.682	0.549	0.134
48	success	0.716	0.656	0.059
49	success	0.573	0.594	-0.022
117	success	0.703	0.593	0.110
126	success	0.755	0.572	0.182
Mean		0.700	0.595	0.105

C DOWNSTREAM CONFIRMATORY DETAILS

C.1 FIXED-STATE ACTION DISTRIBUTIONS

The distributional action stress test retains all four intervention states from the action-reach experiment and uses eight rollouts per memory condition. The per-state success-memory versus failure-memory TV distances are 0.625 for task 22 and 0.375 for task 23. They are 0.125 for task 24 and 0.250 for task 26. Their mean is 0.34375. The frozen 100,000-draw permutation audit gives $p = 0.310527$.

C.2 FROZEN TERMINAL MATRIX

The terminal confirmatory experiment reuses every complete F0 pair and every future task frozen for the earlier fixed-evidence run. Source-memory tasks form $\{21, 22, 23, 25\}$. Future tasks form $\{164, 385, 387, 388\}$. The design therefore contains 16 source–future cells. Each cell receives eight fresh success-memory rollouts and eight fresh failure-memory rollouts. The policy model is Doubao Seed 2.0 Mini at temperature 0.2. All 256 requested calls complete with zero provider failures.

The primary statistic is the mean across cells of the absolute difference between success-memory and failure-memory terminal benchmark success rates. The support gate is frozen before calls at an effect floor of 0.15 together with a within-cell permutation $p < 0.05$. The permutation audit uses 100,000 draws with seed 20260822. The observed statistic is 0.15625 and the audit gives $p = 0.00074$.

C.3 HETEROGENEITY AND CORRUPTION-RATE DECOMPOSITION

For each frozen cell, the confirmatory run estimates paired conditional success probabilities. Eight of the 16 cells have zero observed success-rate contrast. Six have positive failure-memory minus success-memory effects and two have negative effects. The largest squared effects occur at source/future pairs 22/388 and 25/387; together they contribute 78.2051% of the total squared-effect mass. This sign and magnitude heterogeneity is why the revised paper treats the experiment as a forced-leverage measurement rather than a uniform directional effect or a native end-to-end transport estimate.

If reward-label corruption selects the failure-conditioned memory with probability q, the induced between-memory variance equals $q(1-q)\Delta_c^2$ in cell c. The exact mean squared conditional effect

Table 3: F2R1 confirmatory terminal success rates. S denotes success-label memory and F denotes failure-label memory.

Source	Future	S rate	F rate	$\lvert\Delta\rvert$
21	164	1.000	1.000	0.000
21	385	0.000	0.250	0.250
21	387	0.125	0.250	0.125
21	388	1.000	0.625	0.375
22	164	1.000	1.000	0.000
22	385	0.000	0.125	0.125
22	387	0.125	0.000	0.125
22	388	0.250	1.000	0.750
23	164	1.000	1.000	0.000
23	385	0.000	0.000	0.000
23	387	0.125	0.250	0.125
23	388	1.000	1.000	0.000
25	164	1.000	1.000	0.000
25	385	0.000	0.000	0.000
25	387	0.125	0.750	0.625
25	388	1.000	1.000	0.000
	Mean absolute difference			0.15625

across all 16 cells is 0.076172. Averaging over cells gives the curve $0.076172q(1-q)$, with value 0.006855 at $q = 0.1$, value 0.014282 at $q = 0.25$, and value 0.019043 at $q = 0.5$. This remains a descriptive plug-in calculation conditional on forced memory exposure; it is not used as the main claim in the revised stage-resolved paper.

C.4 Native transport audit added after the forced-leverage study

The later frozen native-pipeline audit changes the interpretation of the forced matrix. In Shopping, branch-specific source memory is retrieved in 125/172 held-out opportunities after bank expansion, but the mean absolute native terminal success difference is only 0.02083 with permutation $p = 0.4289$; 34/36 terminal cells are zero. The corresponding first-action total-variation statistic is 0.06944 with $p = 0.5801$, and no frozen state changes its modal first action between reward branches. A post-hoc working-memory branch-shift diagnostic is 0.00335 with $p = 0.2052$.

The cross-domain Reddit audit contains four complete success/failure writes, all of which diverge. Its native terminal mean absolute difference is 0.125 with $p = 0.2253$; 6/8 cells are zero and the two nonzero cells have opposite signs. These later measurements are why the main text separates forced leverage from native transport and treats attenuation after exposure as the central mechanism boundary.

For finite-cell sensitivity, we run a fixed-seed nonparametric bootstrap with 100,000 repetitions, resampling 16 source–future cells with replacement. The percentile 95% interval for mean absolute effect is $[0.054688, 0.273438]$. The corresponding interval for mean squared effect is $[0.010742, 0.164062]$, which maps to $[0.002686, 0.041016]$ for $V(0.5)$. This bootstrap is descriptive sensitivity to the observed cell composition. The preregistered within-cell permutation test on the 256 confirmatory rollouts supplies the primary significance result.